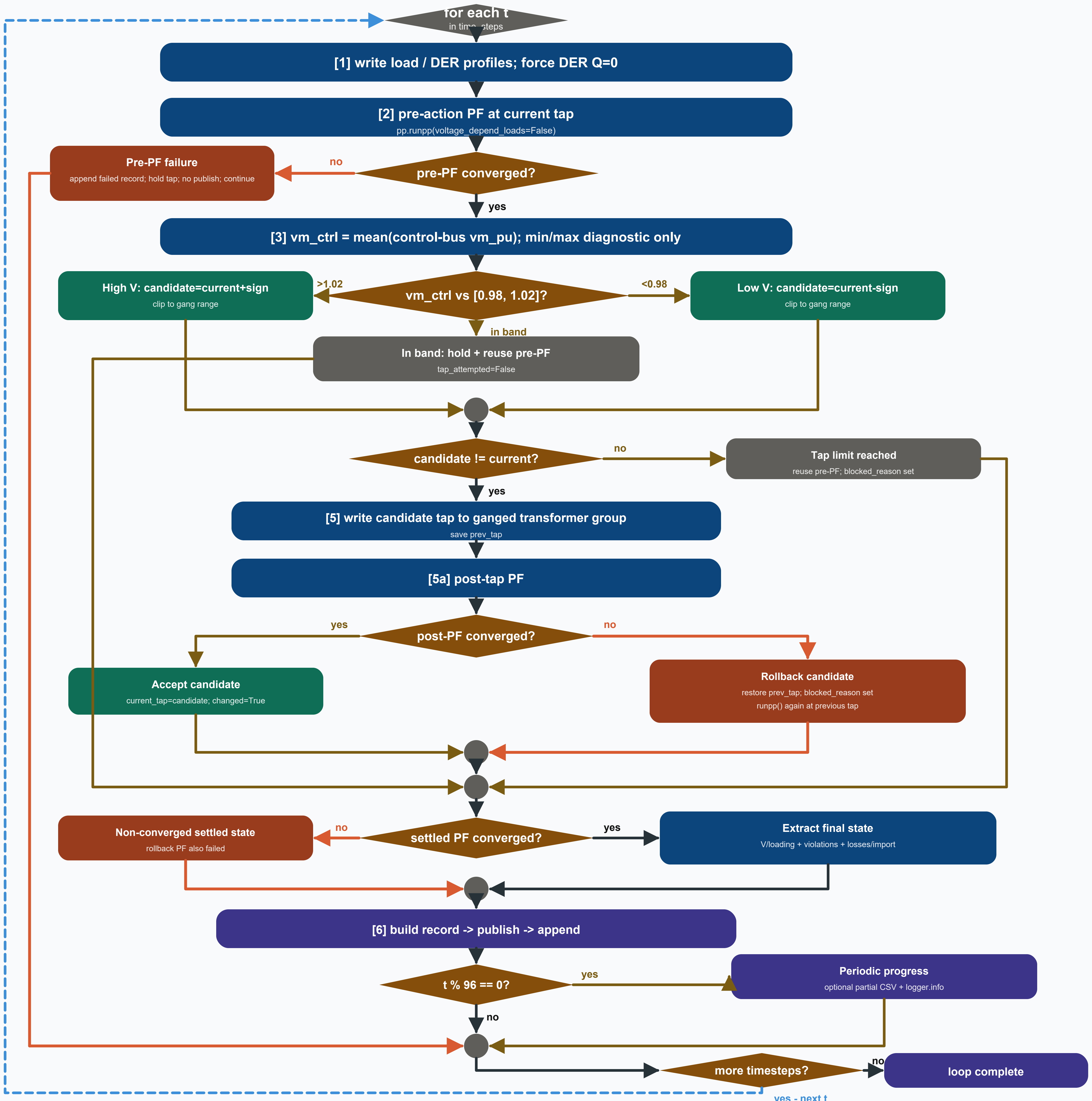

OLTC decision semantics

Control signal is the mean voltage across all selected secondary busbars.

vm_ctrl > 1.02: candidate = current_tap + calibrated sign.

vm_ctrl < 0.98: candidate = current_tap - calibrated sign.

Only one tap step is requested per timestep; candidate is clipped to the ganged range.

Inside the deadband no post-PF is run; valid pre-PF results are reused.

If clipping returns current_tap, the controller records tap_limit_reached and also reuses pre-PF.

Post-tap + rollback semantics

Candidate PF converges: accept the move and set tap_changed=True.

Candidate PF fails: restore prev_tap and set blocked_reason=post_pf_non_convergence.

A second runpp() is attempted after rollback to rebuild valid result tables.

Rollback success is recorded as converged=True but post_pf_reused=False.

Rollback failure leaves converged_post=False and empty result tables.

No reactive-power control or active-power curtailment exists in Scenario 2.

Publisher / failure semantics + implementation note

Normal path: build record -> on_timestep(rec) -> append -> optional t%96 progress.

Pre-PF failure appends its failed record directly, then continue: no on_timestep() and no t%96 branch.

The pre-PF failure rail and blue next-timestep rail are intentionally separate.

Current source edge case: if rollback PF also fails, der_gen_mw_t/load_mw_t are not assigned in that branch.

If this is the first such occurrence record construction can raise; otherwise stale prior-loop values can be reused.

This note reflects the supplied implementation and is not silently corrected by the diagram.